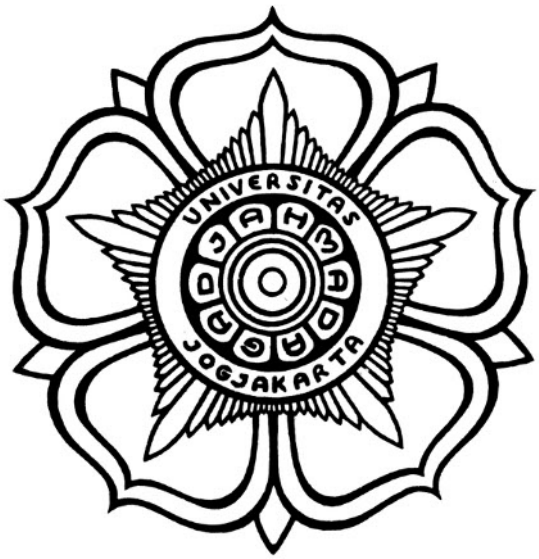
Comparison of *Linear Discriminant Analysis* (LDA) and *Quadratic Discriminant Analysis* (QDA) Models for the Classification of Malignant and Benign Breast Cancer Cells



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1 Introduction

Cancer is one of the leading causes of death worldwide. The *World Health Organization* (WHO) reported that in 2012, the five most common types of cancer found in men worldwide were lung, prostate, colorectal, stomach, and liver cancer, while in women the most common cases were breast, colorectal, lung, cervical, and stomach cancer.

In 2020, the *World Health Organization* (WHO) recorded that 2.3 million women were diagnosed with breast cancer, resulting in approximately 685,000 deaths globally. By the end of 2020, 7.8 million women had been diagnosed with breast cancer within the previous five years, making it the most common cancer in the world.

Breast cancer is a cancerous condition that arises in the breast area. According to the *World Health Organization* (WHO), breast cancer consists of abnormal breast cells that grow uncontrollably and form a tumor. Meanwhile, according to the Indonesian Ministry of Health, breast cancer is a malignant tumor formed from breast cells that grow and develop uncontrollably, spreading to nearby tissues or organs, or to other parts of the body.

In general, the progression of an abnormal breast condition into cancer cells is divided into two classes: benign (*benign*) and malignant (*malignant*) tumors. A tumor itself is an abnormal mass of tissue. A benign tumor is a non-cancerous abnormal cell growth that is not aggressive toward surrounding tissue, whereas a malignant tumor, or cancer, consists of highly aggressive cells that spread and damage nearby tissues and organs.

Early detection of this disease can help prevent an increase in mortality. Therefore, *machine learning* is used to analyze the data and build a classification model aimed at determining whether a given cancer cell is benign or malignant. This *project* uses the dataset linked here. The dataset consists of 569 records with 32 parameters; however, only 8 parameters are used here for efficiency.

2 Theoretical Background

In model evaluation, there are two important observations for determining the model. These observations are made by examining the *confusion matrix* and the ROC curve of the given model. Definitions of these are provided below.

2.1 Data Structure, Essential Metrics, and *Confusion Matrix*

When evaluating a classification model using *testing* data, the *confusion matrix* provides a clear visualization. The *confusion matrix* presents information on the number of test results in terms of *True Positive* (TP), *True Negative* (TN), *False Positive* (FP), and *False Negative* (FN) for the model used. Definitions of these four components are given below.

True Positive: data correctly predicted as positive (1).

True Negative: data correctly predicted as negative (0).

False Positive: data that is actually negative (0) but incorrectly predicted as positive (1).

False Negative: data that is actually positive (1) but incorrectly predicted as negative (0).

In addition, the essential metrics used in evaluation are defined as follows.

Accuracy: the ratio of correct predictions to the total number of predictions made. When the classes are imbalanced, this metric can be *misleading*.

Precision: the ratio of *true positive* predictions to all positive predictions. This metric is prioritized when the consequences of a *false positive* prediction are severe.

Recall/Sensitivity: the ratio of positive predictions to the actual positive cases. This metric is prioritized when the consequences of failing to predict an actual positive case are severe.

Specificity: the ratio of negative predictions to the actual negative cases. This metric is essential when the modeling objective is to accurately identify negative cases.

F1-Score: this metric represents the average of *precision* and *recall*.

The structure of the *confusion matrix* and the essential metrics mentioned above, along with their formulas, are shown in the figure below.

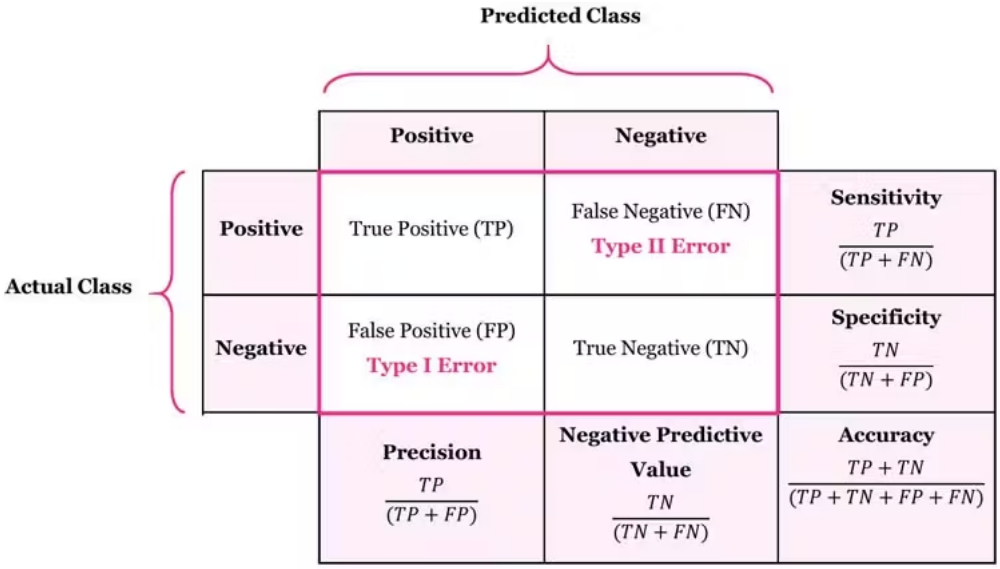


Figure 1: *Confusion Matrix* and Essential Metrics (source: Encord)

2.2 ROC Curve and AUC Value

- **ROC**: The ROC curve (Receiver Operating Characteristic curve) is a graph showing the performance of a classification model across all classification thresholds. This curve plots two parameters
  - **T**rue Positive Rate
  - **F**alse Positive Rate

  The **True Positive Rate (TPR)** is synonymous with recall and is therefore defined as follows:

$$TPR = \frac{TP}{TP + FN}$$

The **False Positive Rate (FPR)** is defined as follows:

$$FPR = \frac{FP}{FP + TN}$$

The ROC curve plots TPR versus FPR at different classification thresholds. Lowering the classification threshold classifies more items as positive, thereby increasing both *False Positive* and *True Positive* counts. The following figure shows a standard ROC curve.

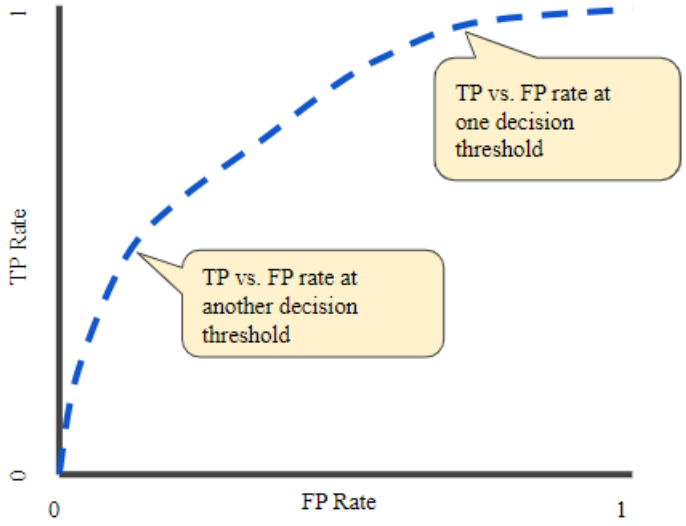


Figure 2: Ratio of TP vs. FP at Different Classification Thresholds.

To compute points on the ROC curve, one could evaluate the model repeatedly at different classification thresholds, but this would be inefficient. Fortunately, there is an efficient sorting-based algorithm that provides this information, called AUC.

- **AUC**: stands for *Area Under Curve*. This means AUC measures the entire two-dimensional area beneath the ROC curve, from (0,0) to (1,1), as shown in the figure below. AUC provides an aggregate measure of performance across all possible classification thresholds. One way to interpret AUC is as the probability that the model ranks a random positive example higher than a random

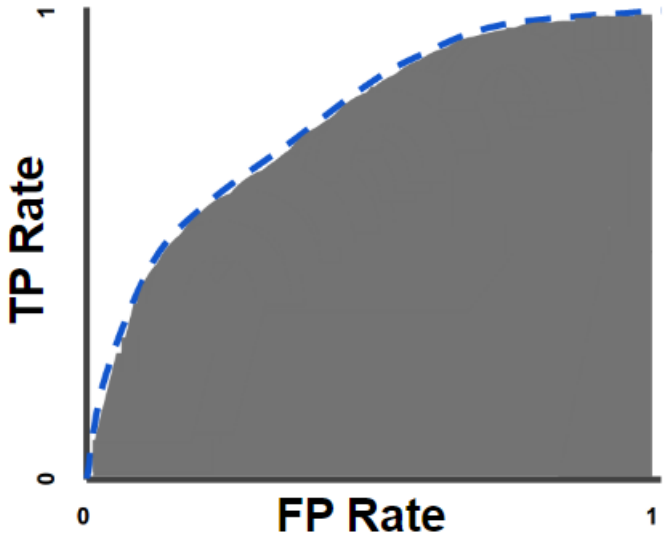


Figure 3: AUC (Area Under the ROC Curve).

negative example. For instance, consider the following example, arranged from left to right in increasing order of logistic regression prediction score:

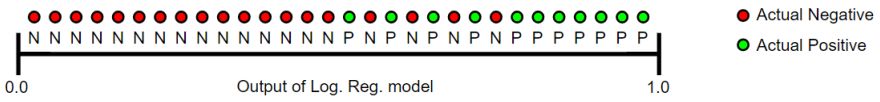


Figure 4: Predictions Sorted by Increasing Logistic Regression Score.

AUC represents the probability that a random positive example (green) is positioned to the right of a random negative example (red).
AUC values range from 0 to 1. A model whose predictions are 100% wrong has an AUC of 0.0; a model whose predictions are 100% correct has an AUC of 1.0.
AUC is desirable for the following two reasons:

- **AUC is scale-invariant**. It measures how well predictions are ranked, rather than their absolute values.
- **AUC is classification-threshold-invariant**. This metric measures the quality of the model's predictions, regardless of the chosen classification threshold.

However, both of these properties come with caveats that can limit the usefulness of AUC in certain use cases:

- **Scale invariance is not always desirable**. For example, well-calibrated probability outputs are sometimes essential, and AUC does not provide information about calibration.
- **Classification-threshold invariance is not always desirable**. When there is a significant difference in the consequences of *false negatives* versus *false positives*, it becomes critical to minimize one particular type of classification error. For example, in email spam detection, one might want to prioritize minimizing *false positives* (even if this causes a significant increase in *true negatives*). AUC is not a useful metric for this kind of optimization.

3 Data Analysis

Before the data analysis is carried out, the available data is processed through computations in Google Colab, accessible via the following link.

3.1 Parameter Identification

The data used consists of ten characteristics of breast cancer. It contains 10 variables/parameters, comprising 2 control parameters and 8 variation parameters. The parameters used are as follows.

1. id, the identification number of the cancer diagnosis.
2. diagnosis, the diagnosis of the cancer cell: M for *Malignant* and B for *Benign*.
3. radius_mean, the average distance from the center to points on the boundary of the cancer cell.
4. texture_mean, the average surface texture of the breast.
5. perimeter_mean, the average outer boundary length of the cancer cell.
6. area_mean, the average area of the breast lobule.
7. smoothness_mean, the average smoothness level of the cancer cell boundary.
8. compactness_mean, the average density of the breast.
9. concavity_mean, the average severity of concave portions of the breast contour.
10. concave_points_mean, the average number of concave points on the boundary of the cancer cell nucleus within a concavity of the breast.

3.2 Data Cleaning

The data is provided in two groups: a model development group consisting of 569 records, and a validation group obtained from the training data results. Data cleaning was not performed, since the data obtained was already in good condition, with no incomplete columns. The results of checking the model-building data are shown in the code output below.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
```

```
Data columns (total 10 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   id                   569 non-null    int64
 1   diagnosis            569 non-null    int64
 2   radius_mean          569 non-null    float64
 3   texture_mean         569 non-null    float64
 4   perimeter_mean       569 non-null    float64
 5   area_mean            569 non-null    float64
 6   smoothness_mean      569 non-null    float64
 7   compactness_mean     569 non-null    float64
 8   concavity_mean       569 non-null    float64
 9   concave points_mean  569 non-null    float64
dtypes: float64(8), int64(2)
memory usage: 44.6 KB
```

3.3 Exploratory Data Analysis

3.3.1 Observation of Variable Distribution

Histograms are generated for all parameters used, excluding the id and diagnosis parameters since these are not numerical. The resulting figure is shown below.

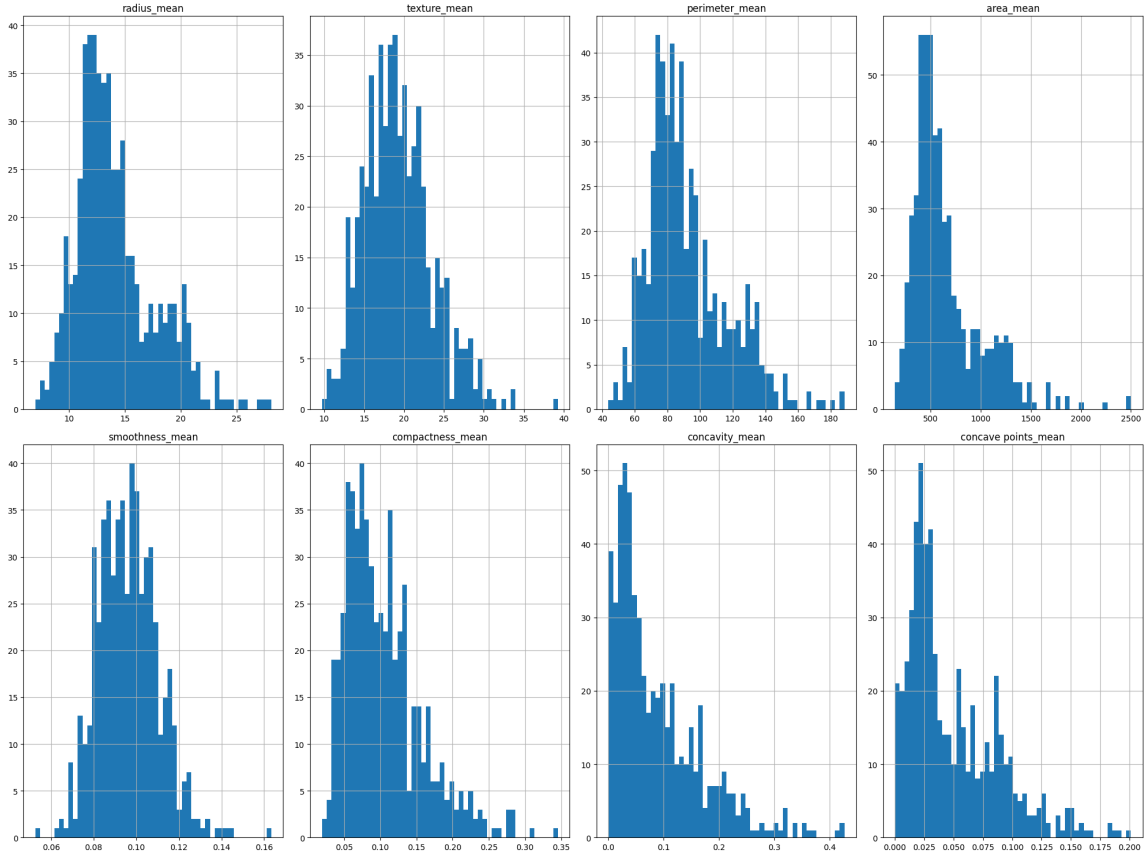


Figure 5: Distribution of Parameter Data.

Of the 8 parameters, it can be seen that only `radius_mean` and `texture_mean` reasonably approximate a normal distribution, while the others more closely approximate a chi-square distribution.

3.3.2 *Feature Selection*

Variable selection is performed to determine which variables will be used in this classification model. Variables are selected based on their influence on the `diagnosis` variable, assessed via a one-way ANOVA test or a Kruskal-Wallis test. The remaining variables are then tested for their correlation with the `diagnosis` variable, with a variable considered sufficiently good if its correlation value is greater than 0.1.

With $\alpha = 0.05$, a normality test is performed to check whether the data can be tested using parametric methods. The normality test is conducted using D'Agostino's K-squared test, with the null hypothesis that the variable is normally distributed. Under this test, all variables fail the normality test, yielding p-values smaller than α. Since parametric testing cannot be applied to any of the variables, a nonparametric test is used instead for variable selection.

A Kruskal-Wallis test is performed for each variable against the `diagnosis` variable, replacing the one-way ANOVA test, which cannot be used since none of the data is normally distributed. Taking $\alpha = 0.05$ and the null hypothesis that the variable used has no effect on changes in the `diagnosis` variable, the Kruskal-Wallis test — implemented via the `stats.kruskal` feature from the `scipy` library in Python — yields no p-value greater than α, indicating that all parameters are significant.

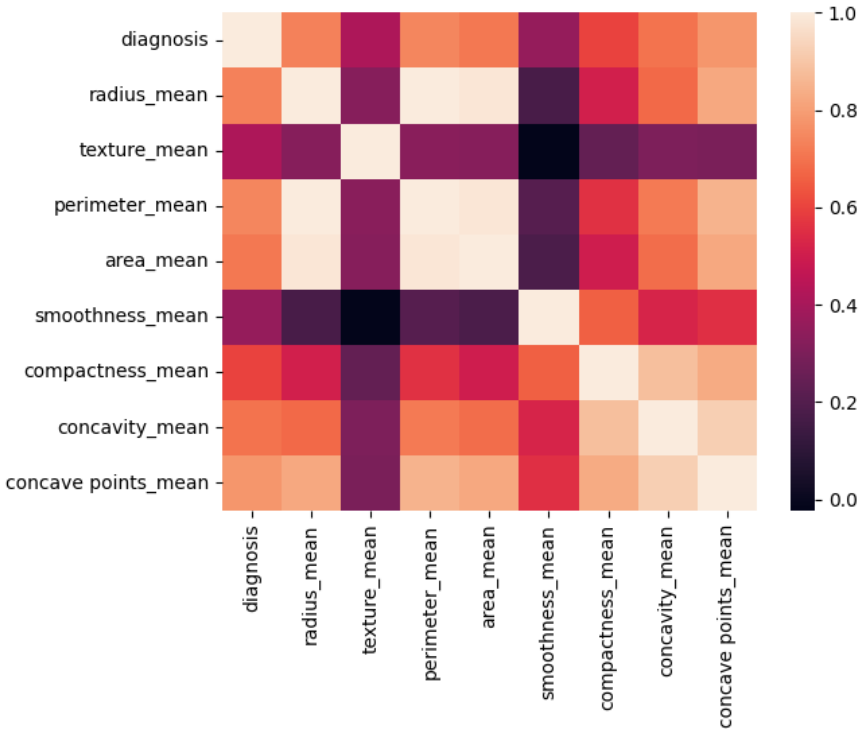


Figure 6: Correlation *Heatmap* Between Variables

From Figure 6, two variables show a low correlation with the `diagnosis` variable — that is, a correlation below 0.1 — namely `texture_mean` and `smoothness_mean`. These two variables are therefore excluded, leaving 6 strongly influential variables to be used in this model.

4 Classification Models

Four models were considered in determining the best classification model for data validation, namely *linear discriminant analysis* (LDA) and *quadratic discriminant analysis* (QDA), using the *default* threshold of 0.5.

Model construction is carried out using features from the `sklearn` library for Python. In building the models, 75% of the 569-record dataset is used as *training* data, and the remainder is used as *testing* data. To evaluate the resulting models, a classification *report*, *confusion matrix*, and ROC curve with AUC value are presented. The best model is selected based on accuracy, *sensitivity*, and the highest AUC value. In the output provided for this evaluation, the *precision* value for the negative class (0) corresponds to the *negative predictive value* in the *confusion matrix*, and the *precision* value for the positive class (1) corresponds to the *precision* value in the *confusion matrix*. Likewise, the *recall* value for the negative class (0) corresponds to the *specificity* in the *confusion matrix*, and the *recall* value for the positive class (1) corresponds to the *sensitivity* in the *confusion matrix*.

4.1 Justification of *Likelihood-Ratio*-Based Models

LDA and QDA can be viewed as *likelihood ratio*-based methods, since their underlying distribution is a multivariate Gaussian. LDA assumes equal covariance across classes, while QDA makes no such assumption. Both methods estimate the class distributions. To build a classifier, they then compute a line (LDA) or a quadratic curve (QDA) corresponding to a naive Bayesian classifier based on likelihood or on the posterior (when prior class probabilities are included). This line or curve passes through the points where both classes have equal likelihood.

4.2 *Linear Discriminant Analysis* (LDA) Model

The graphical results from *testing* the LDA model are as follows.

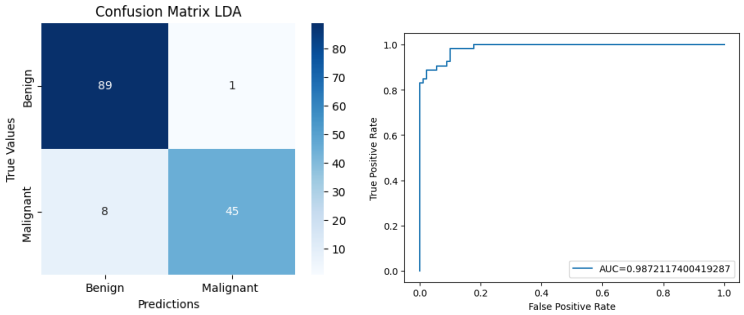


Figure 7: Testing of the *Linear Discriminant Analysis* Model

The classification *report* is presented below.

```
Accuracy Score : 0.9230769230769231
Classification Report
              precision    recall  f1-score   support

           0       0.94      0.93      0.94        90
           1       0.89      0.91      0.90        53

    accuracy                           0.92       143
   macro avg       0.92      0.92      0.92       143
weighted avg       0.92      0.92      0.92       143
```

By examining the information in the ROC curve for the LDA model, the *default* threshold of 0.5 is not the optimal threshold for this model. After experimentation, a threshold of 0.265 was obtained; the *confusion matrix* and classification *report* for the threshold of 0.265 are as follows

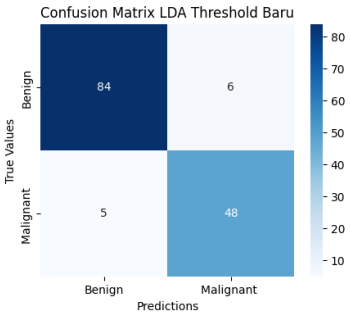


Figure 8: Testing of the *Linear Discriminant Analysis* Model with New Threshold

```
    Accuracy Score : 0.9370629370629371
Classification Report
              precision    recall  f1-score   support

           0       0.92      0.99      0.95        90
           1       0.98      0.85      0.91        53

    accuracy                           0.94       143
   macro avg       0.95      0.92      0.93       143
weighted avg       0.94      0.94      0.94       143
```

4.3 *Quadratic Discriminant Analysis* (QDA) Model

The graphical results from *testing* the QDA model are as follows.

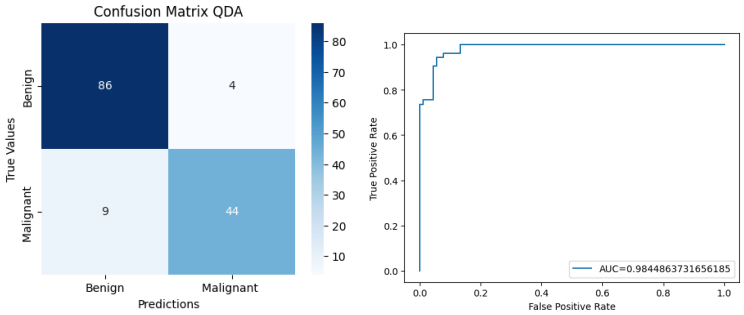


Figure 9: Testing of the Logistic Regression Model

The classification *report* is presented below.

```
Accuracy Score : 0.9090909090909091
Classification Report
              precision    recall  f1-score   support

           0       0.91      0.96      0.93        90
           1       0.92      0.83      0.87        53

    accuracy                           0.91       143
   macro avg       0.91      0.89      0.90       143
weighted avg       0.91      0.91      0.91       143
```

4.4 Best Model Selection and Interpretation

Based on the tests performed, the best AUC, highest accuracy, and highest *sensitivity* for malignant cells were achieved by the *linear discriminant analysis* (LDA) model with a threshold of 0.265. The LDA model is therefore established as the best model among those tested. In addition, the *confusion matrix* and ROC curve for this model can be found in Figure 7 and Figure 8. Based on this model, a breast cancer cell can be classified as malignant if its predicted probability of being a

cancer cell under the LDA model is 0.265 — a relatively small probability is already sufficient to classify a cell as malignant (*malignant*).

5 Results and Conclusion

Based on the selection of the best model, the *linear discriminant analysis* model is used for data validation. The validation data is obtained by turning the earlier training data into testing data, with new training data serving as the validation data, yielding the following breast cancer classification percentages:

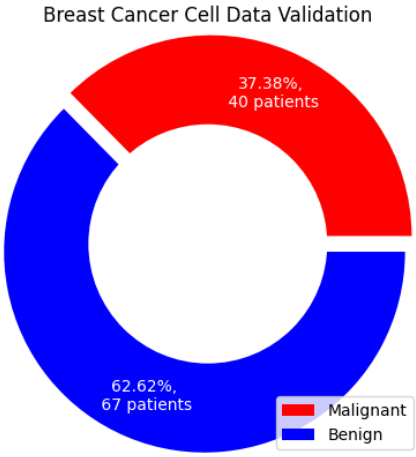


Figure 10: Validation of Breast Cancer Cell Classification Data

From this diagram, it can be seen that 40 out of 107 patients, or approximately 37.38%, are classified as having malignant cancer cells, while 67 out of 107 patients, or approximately 62.62%, are classified as having benign cancer cells.

Therefore, the error rate when using the *linear discriminant analysis* (LDA) model is the smallest, making this model suitable for use.

References

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Appendix

The Google Colab link for the calculations can be accessed at https://colab.research.google.com/drive/1xRZEW0R_diGhcMZ9Vzff7cQIhFYNzkaC?usp=sharing